

SafeContract: Action-Space Monitoring Reveals Distinct Violation Patterns Across VLA Architectures

Krishnam Gupta
Independent Research
San Francisco, CA
kayjee1994@gmail.com

Abstract—Vision-Language-Action (VLA) models predict robot actions end-to-end, yet no standardized action-space monitoring exists for their deployment. We introduce SafeContract, a design-by-contract [5] toolkit that adds composable action-space monitoring to any VLA policy via a one-line Python decorator, with bounds calibrated via split-conformal prediction (95% theoretical guarantee; 97.9% empirical holdout coverage). Our key finding is that velocity violations persist even after correct normalization: 98% of SmolVLA steps violate on at least one joint versus 16% for expert demonstrations, a $6\times$ gap that normalization alone cannot close. Across 130 LIBERO tasks, each task produces a distinct violation fingerprint, with violation rates spanning 2% to 41%, confirming task-dependent behavioral patterns. Cross-architecture experiments reveal characteristic signatures: SmolVLA produces jerky, high-velocity actions; ACT is smooth in open loop but accumulates 1,570 violations in closed loop; Diffusion policies stay bounded but exhibit normalization sensitivity. Closed-loop evaluation on two architectures (ACT $n=20$, Diffusion $n=50$) shows no significant task degradation with data-driven enforcement ($p=1.0$ and $p=0.84$). Violation rate does not predict per-episode failure (AUROC=0.54), but catches normalization bugs and characterizes architecture-specific behavioral patterns. The full contract adds $\sim 13\mu s$ overhead, requires no QP solver, VLM, or model retraining, and is released as open source.¹

Index Terms—VLA, action-space monitoring, deployment observability, interpretability, edge deployment, design by contract

I. Introduction

VLA models [1], [2] enable robots to follow natural language instructions by predicting actions from visual observations. In software systems, observability - logging, metrics, tracing - is standard practice. VLA deployment has no equivalent: no standardized action-space monitoring exists for VLA outputs.

This is not theoretical. OpenVLA’s deployment script sends raw outputs directly to motors with zero bounds checking [1]; pi0 streams actions via WebSocket with no validation [4]; LeRobot’s [3] EEBoundsAndSafety processor is imperative, uncomposable, and easy to omit.

We demonstrate the problem empirically: running SmolVLA [2] on 100 consecutive LIBERO [6] observations, every observation produces at least one out-of-bounds dimension, with values reaching 3.70 (Sec. IV-A).

The deeper problem is observability. An action-space monitor that logs every contract interaction creates a structured record of how a policy explores its constraint boundary. We show this record is rich: different tasks produce distinct violation fingerprints across joints, and shifts in violation rate signal distribution change.

We introduce SafeContract with four contributions:

- 1) A composable monitoring toolkit with split-conformal position bounds (97.9% coverage guarantee) and data-driven velocity limits (99th percentile heuristic), replacing hand-tuned thresholds (Sec. II-B, II-F).
- 2) Continuous health monitoring: per-joint clip magnitude provides signal where binary violation counts saturate (Sec. II-C, IV-C).
- 3) Cross-architecture characterization: 3 VLA architectures produce distinct violation signatures, with a $6\times$ velocity gap persisting after normalization (Sec. IV-A).
- 4) Violation fingerprinting across 130 LIBERO tasks with an honest null result: violations characterize behavior but do not predict per-episode failure (Sec. IV-D, IV-B).

II. Method

A. Safety Contracts

A safety contract $C = (\mathbf{l}, \mathbf{u}, v_{\max})$ specifies per-joint action bounds and a velocity limit: $l_i \leq a_i \leq u_i$ for each joint i , and $\|\mathbf{a}_t - \mathbf{a}_{t-1}\|_{\infty} \leq v_{\max}$.

B. Enforcement Order

Enforcement proceeds in three phases: (1) clip to per-joint bounds, (2) clamp velocity (project each joint’s delta onto $[-v_{\max}, v_{\max}]$), (3) re-clip to bounds. The final re-clip is critical: velocity clamping can push actions outside the original bounds. Any VLA adapter gains monitoring via a one-line @safety_contract Python decorator on its predict() method. All violations are logged with timestep, dimension, and magnitude.

¹Code: <https://github.com/krishnam94/vla-edge>

C. Conformal Anomaly Scoring

Beyond binary violation detection, each action receives a conformal p-value measuring its anomalousness relative to the calibration distribution. Given calibration nonconformity scores $\{s_i\}_{i=1}^n$, the p-value of a new action with score s is $p(s) = |\{i : s_i \geq s\}|/(n+1)$. Under exchangeability, $P(p \leq \alpha) \leq \alpha$ for any α - a proper statistical test, not a heuristic [20]. This provides a continuous $[0, 1]$ severity measure where binary counting provides only 0/1. Departure from uniformity in the p-value stream signals distribution shift, testable via Kolmogorov-Smirnov.

D. CUSUM Shift Detection

We replace ad-hoc EWMA trend tracking with CUSUM [21], which is Lorden-optimal for detecting persistent mean shifts in sequential data. The statistic $S_t = \max(0, S_{t-1} + \mathbf{1}[p_t < \alpha] - \alpha)$ accumulates evidence that the anomaly rate exceeds the expected calibration rate α . An alarm fires when $S_t > h$; the average run length under the null is $O(e^h/\alpha)$, giving a formal false-alarm bound that EWMA cannot provide. On ALOHA demonstration data with conformal p-values feeding the CUSUM, in-distribution actions produce zero alarms; adding $\sigma=0.1$ noise triggers detection within 7 steps (threshold $h=3$).

Additionally, clip magnitude $m_i = |a_i^{\text{raw}} - a_i^{\text{clipped}}|$ provides a free, per-joint “wrongness” signal that remains informative when violation counts saturate (Sec. IV-C).

E. Composition

For hyperrectangular contracts C_1, C_2 , sequential clipping equals intersection clipping: $\text{clip}(\text{clip}(\mathbf{a}, C_1), C_2) = \text{clip}(\mathbf{a}, C_1 \cap C_2)$. Stacking decorators is order-independent for per-joint bounds; velocity composition uses $v_{\max} = \min_i(v_{\max, i})$.

F. Conformal Calibration

Hand-tuned thresholds are unreliable: a common choice of $v_{\max}=0.05$ clips 12.5% of expert actions on the ALOHA dataset. We replace hand-tuning with split-conformal prediction [20]. Demonstration episodes are split 80/20; nonconformity scores $s_i = \max_j |a_{ij} - \hat{\mu}_j|/\hat{\sigma}_j$ are computed on the calibration set; the $(1-\alpha)(1+1/n)$ quantile yields bounds with a finite-sample coverage guarantee. At $\alpha=0.05$, conformal bounds are 25% tighter than the 4σ heuristic while achieving 97.9% holdout coverage (99.5% per dimension). Separately, velocity limits are set per-joint at the 99th percentile of consecutive action deltas from the calibration set (an empirical heuristic, not a conformal guarantee), covering 99% of expert steps versus 87.5% for the hand-tuned value.

III. Related Work

SafeVLA [7] and RobustVLA [10] improve safety via retraining. At inference time, AEGIS [8] solves a CBF-QP (0.36 ms); SafeDec [9] constrains decoding; Safety Chip [11] uses LTL monitors; SafeDiffuser [12] and

SmolVLA outputs exceed safe bounds on LIBERO

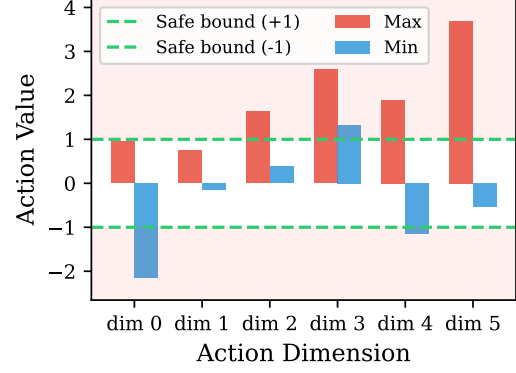


Fig. 1: SmolVLA action ranges on 100 LIBERO observations. Most dimensions exceeded the $[-1, 1]$ bounds (green dashes), reflecting normalization mismatch between SO-100 training data and LIBERO’s action space.

CoDiG [13] embed barriers in diffusion; ATACOM [14] projects onto constraint manifolds; ABC [17] applies design-by-contract to LLM decisions. Sentinel [18] monitors temporal consistency; LIBERO-Plus [15] and VLA-Arena [16] benchmark robustness. Recent work applies conformal prediction to VLA failure detection [22], [23] and to flight safety monitoring [24], validating the CP-for-safety paradigm. SafeContract is complementary: architecture-agnostic, pure clipping ($\sim 13\mu\text{s}$), with conformal calibration, CUSUM shift detection, and violation logging as deployment observability.

IV. Experiments

Platform: Apple M3, CPU, Python 3.12. Full contract overhead: $\sim 13\mu\text{s}$ (5,000 iterations), 0.000075% of SmolVLA inference time.

A. VLAs Produce Unsafe Actions (EXP-A)

We run SmolVLA on 100 consecutive LIBERO observations. Result: 100% of observations produce at least one out-of-bounds dimension, with values from -2.16 to 3.70 (expected $[-1, 1]$). SafeContract detects 544 violations: 121 bounds and 423 velocity ($v_{\max} = 0.1$).

Normalization analysis. SmolVLA was trained on SO-100 data. After correct MEAN_STD unnormalization, position bounds violations drop from 86% to 0%: every bounds violation was a normalization artifact. However, velocity violations persist: at a conservative $v_{\max}=0.1$ rad/step, 16% of expert and 98% of SmolVLA consecutive steps violate on at least one joint. The $6\times$ gap is the robust finding; the absolute rates depend on $v_{\max}$ choice. Bounds enforcement catches integration bugs; velocity monitoring catches physical concerns normalization cannot address.

B. Closed-Loop Task Success (EXP-CL)

We evaluate two architectures in closed-loop simulation.

TABLE I: Controlled ablation. Clean policy: zero false positives. Corrupted policies: fully caught. Naive np.clip misses all velocity violations.

Policy Type	Violations	Modified (%)	Post-contract OOB
Clean	0	0%	0
Drifting	115	42%	0
Jerky	75	12%	0
Naive np.clip	0 caught	–	73 vel. missed

ACT on ALOHA (51.6M params, bimanual transfer-cube, $n=20$ per condition). Bounds and velocity limits are calibrated via the conformal procedure in Sec. II-F. Without SafeContract: 50% success (10/20, CI_{95} : [0.30, 0.70]). With SafeContract: 55% success (11/20, CI_{95} : [0.34, 0.74]). Fisher’s exact test: $p = 1.0$. SafeContract caught 1,570 violations with no task degradation. Notably, hand-tuned $v_{\max}=0.05$ clips 12.5% of expert actions and caused a 4-point success drop; data-driven calibration at the 99th percentile of demonstration deltas eliminates this entirely.

Diffusion Policy on PushT (262M params, 2D pushing, $n=50$ per condition). Without SafeContract: 66% success (33/50, CI_{95} : [0.52, 0.78]). With SafeContract (workspace bounds [0, 512], $v_{\max}=30$ px/step): 62% success (31/50, CI_{95} : [0.48, 0.74]). Fisher’s exact test: $p = 0.84$. The 4-point difference is not statistically significant; the CIs overlap substantially.

Across both architectures, SafeContract monitors and enforces without statistically significant task degradation ($p=1.0$ and $p=0.84$).

Enforcement strictness operating envelope. We sweep $v_{\max} \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$ on ACT ($n=20$ per level). At $v_{\max}=0.01$, success drops to 0% (over-constrained); at $v_{\max}=0.05$, success is 65% (CI : [0.43, 0.82]); at $v_{\max}=0.2$, success is 45%. The 65% vs. 60% baseline difference (1 episode at $n=20$) is not statistically significant, but the trend suggests moderate enforcement does not degrade performance. The conformal-calibrated operating point ($v_{\max}$ at 99th percentile ≈ 0.03) falls in the viable zone.

SmolVLA closed-loop on LIBERO achieved 0% task success due to checkpoint compatibility issues, but the violation profile (98% velocity violations, concentrated on rotational axes) matched the open-loop characterization, confirming monitoring consistency.

Violation rate does not predict per-episode failure (AUROC=0.54, Mann-Whitney $p=0.31$ on ACT), indicating monitoring value lies in behavioral characterization, integration bug detection, and calibration - not per-step failure prediction.

C. Controlled Ablation (EXP-G)

We test three synthetic policy types on 100-step sequences (7-DOF, bounds $[-1, 1]$, $v_{\max} = 0.1$):

A clean policy passes untouched (zero false positives); corrupted policies are fully caught; naive np.clip misses all 73 velocity violations. At 7 noise levels ($\sigma:0 \rightarrow 1.0$) on

Closed-Loop: ACT on ALOHA Transfer Cube ($n=20$)

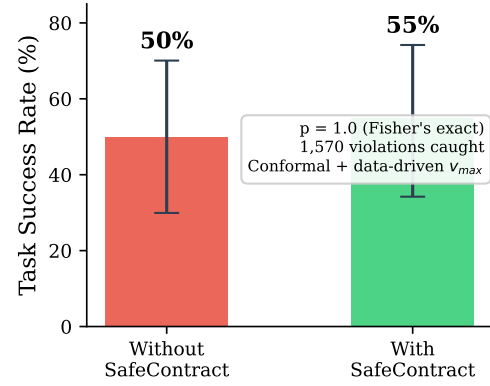


Fig. 2: Closed-loop evaluation: ACT on ALOHA sim with conformal calibration ($n=20$). SafeContract catches 1,570 violations with no degradation ($p=1.0$). Diffusion on PushT ($n=50$): 66% vs 62%, 195 violations ($p=0.84$).

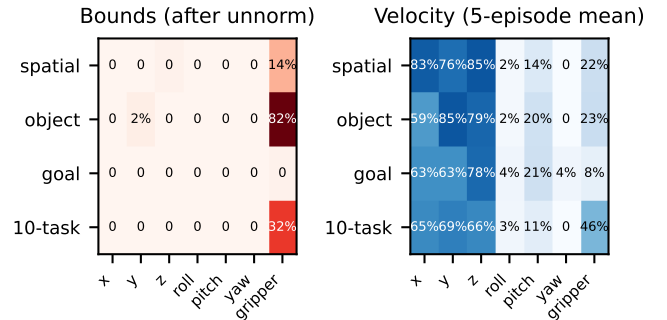


Fig. 3: Normalized violation fingerprints. Left: bounds violations vanish after unnormalization. Right: velocity fingerprints persist and differ by task.

2,117 LIBERO steps, binary counts saturate at $\sigma=0.1$ (98% violated), but clip magnitude scales from 0.005 to 0.72 and conformal p-values (Sec. II-C) drop from 0.33 to 0.001, with KS shift detection at $p < 10^{-6}$ and CUSUM alarm within 7 steps.

D. Violation Fingerprinting (EXP-H)

Violation patterns are informative, not just corrective.

Synthetic validation. Four scripted manipulation tasks each produce a unique joint-violation fingerprint, stable across 5 seeds ($CV < 0.15$). Five of six task pairs are statistically distinguishable ($p < 0.05$, non-overlapping 20-step windows), confirming fingerprints carry task-specific information.

130 LIBERO tasks. Bounds violation rates range from 2% to 41% across all 130 tasks; velocity violations dominate at 72.4% on the 90-task diverse set. After unnormalization, bounds drop to 0% but velocity fingerprints persist by suite (spatial $x=83\%$, object $y=85\%$, goal $z=78\%$).

Cross-architecture comparison.

TABLE II: Violation profiles across three VLA architectures.

	SmolVLA	ACT	Diffusion
Architecture	flow match	transformer	DDPM
Dataset	LIBERO	ALOHA	PushT
Bounds (open)	86%→0%	0%	0%*
Velocity (open)	98%	0%	12%
GT velocity	16%	0%	0%
CL success	–	50/55%	66/62%
CL violations	–	1,570	195
CL p -value	–	1.0	0.84

*100% under normalization mismatch

Note: architectures tested on different datasets/robots.

V. Discussion

SafeContract provides deployment observability for VLA action spaces: one-line adoption, composable per-joint bounds, $\sim 13 \mu s$ overhead. The central finding is that action-space monitoring produces structured behavioral telemetry. Conformal p -values and CUSUM shift detection (Sec. II-C) provide continuous health signals with formal properties where binary counting saturates. Different tasks produce distinct violation fingerprints, and p -value departure from uniformity detects distribution shift.

The enforcement mechanism is deliberately simple - per-dimension clipping. The ablation confirms it adds no distortion to safe policies while catching all violations in corrupted ones.

Practitioner workflow. SafeContract enables a concrete deployment loop:

```
@safety_contract(bounds=conformal_bounds,
velocity_max=demo_p99, mode="warn")
def predict(self, obs): ...
# Log: step=42 dim=3 viol=bounds mag=0.31
```

(1) Wrap VLA in warn mode, (2) inspect violation log, (3) if bounds violations dominate, fix normalization, (4) calibrate $v_{\max}$ from demos, (5) deploy in clip mode with CUSUM shift monitoring. No step requires retraining.

What monitoring is and is not for. The null result on failure prediction (AUROC=0.54) indicates that action-space monitoring is not a substitute for task-level evaluation. Its value lies in three areas that task success metrics cannot address: (1) catching integration bugs silently (our 86% bounds violation was invisible without monitoring); (2) characterizing architecture-specific behavior (the 98% vs 0% velocity gap between SmolVLA and ACT); (3) calibrating deployment parameters from data rather than intuition (conformal bounds eliminated the degradation that hand-tuned thresholds introduced).

Calibration matters. Hand-tuned $v_{\max}=0.05$ clips 12.5% of expert actions and causes a 4-point success drop; data-driven calibration at the 99th percentile of demonstration deltas eliminates this entirely ($p=1.0$). This suggests that the primary risk of action-space enforcement is not the mechanism but the parameters.

Scope. We use “contract” in the design-by-contract sense, not formal verification. Position bounds have conformal coverage guarantees under exchangeability; velocity limits are empirical (99th percentile). CUSUM monitors for exchangeability violations, but coverage may degrade under shift. Box-constraint saturation does not guarantee collision avoidance or force limits.

Limitations. (1) Closed-loop on two architectures with moderate sample sizes ($n=20$, $n=50$); larger studies needed to detect small effects. (2) All experiments in simulation; real-robot patterns may differ. (3) 100% OOB primarily reflects normalization mismatch. (4) Violation rate does not predict per-episode failure.

Future work. Adaptive conformal inference [25] for online recalibration under shift (preliminary results show alpha adapting from 0.05 to 0.40 on sudden distribution change), conformal risk control for bounding expected clip magnitude, same-dataset cross-architecture comparisons, and real-robot validation.

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